

# Artificial Intelligence Applications in DWDM Networks: A Systematic Review

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## Abstract

The continuous increase in data traffic driven by the 5G and 6G networks requires a telecommunications infrastructure that is increasingly faster, more dynamic, and more secure. Optical networks based on *Dense Wavelength Division Multiplexing* (DWDM) technology form the fundamental infrastructure of these communications, but they face major challenges at the physical layer and in the complexity of resource management. This paper presents a systematic review of the literature following the PRISMA methodology, analyzing 23 articles on the application of *Artificial Intelligence* (AI) and *Machine Learning* (ML) in DWDM networks. The results indicate that algorithms such as *Artificial Neural Networks* (ANN) and *Support Vector Machines* (SVM) offer accuracies above 90% in predicting the *Quality of Transmission* (QoT), while complex models enable the automation of management and security in quantum channels, namely in *Continuous-Variable Quantum Key Distribution* (CV-QKD) systems. We conclude that the transition to autonomous networks based on *Knowledge-Defined Networking* (KDN) is a fundamental requirement for future-generation infrastructures.

## Index Terms

DWDM, Machine Learning, Deep Learning, Artificial Intelligence, Optical Networks, CV-QKD, Quality of Transmission (QoT), Knowledge-Defined Networking (KDN), 6G.

## I. INTRODUCTION

The expansion of digital services, along with the consolidation of 5G networks and the development of 6G networks, has caused a continuous increase in data traffic [1], [2]. Optical transport networks based on Dense Wavelength Division Multiplexing (DWDM) technology constitute the fundamental infrastructure to sustain this growth, enabling the simultaneous transmission of multiple wavelengths in a single fiber and reaching transmission rates in the order of terabits per second [3].

The operation of these networks at high transmission rates imposes considerable limitations on the physical layer. Phenomena such as Intersymbol Interference (ISI), Four-Wave Mixing (FWM), and various nonlinear distortions compromise signal integrity and require advanced compensation algorithms [4]. Traditional network management, often based on static rules and manual intervention, is insufficient given traffic variability and the demands for latency and reliability [1].

In this context, the application of Artificial Intelligence (AI) techniques offers a solution for automating these networks. The use of Machine Learning (ML) and Deep Learning (DL) models allows for estimating Quality of Transmission (QoT), predicting failures early, and dynamically managing spectral resources [5], [6]. Computational intelligence also increases security, especially in applications that protect quantum channels in Continuous-Variable Quantum Key Distribution (CV-QKD) systems [7].

This report provides a literature review on the application of AI algorithms in DWDM networks. The remainder of the paper is organized as follows: Section 2 details the research methodology, followed by an overview of fundamental theoretical concepts in Section 3. Section 4 analyzes the challenges and opportunities identified in the literature. Finally, Section 5 examines upcoming technological trends, followed by Section 6, which concludes the study.

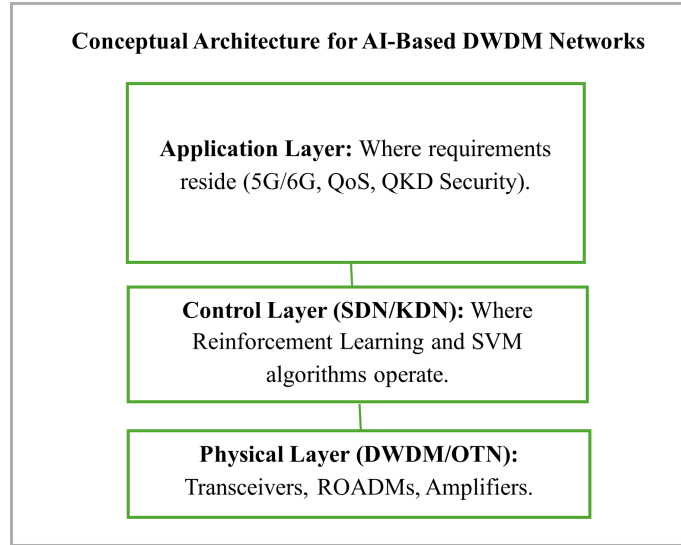


Figure 1. Conceptual Architecture for AI-Based DWDM Networks.

## II. RESEARCH METHODOLOGY

For the literature review, the PRISMA (*Preferred Reporting Items for Systematic Reviews and Meta-Analyses*) method was followed. The article search was conducted across IEEE Xplore, Springer, Wiley, MDPI, Optica, and ResearchGate. The search string used was: ("DWDM" OR "Dense Wavelength Division Multiplexing") AND ("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "AI" OR "Generative AI") AND ("Management" OR "Security" OR "Resilience" OR "Performance" OR "Self-healing" OR "QoT").

The selection of articles was based on three criteria: 1) focus on the optical layer, specifically on DWDM networks or their evolutions, such as Optical Transport Network (OTN); 2) application of artificial intelligence techniques in the solution; and 3) the goal of improving network performance, management, security, or resilience.

In an initial phase, 30 articles were analyzed and selected. Of these 30 articles, 7 were excluded for failing to meet the technological requirements. For example, the studies by Ananda *et al.* [8], Mainas *et al.* [9], and Naskar and Majhi [10] addressed routing, fault protection, and network planning, but used only mathematical algorithms or conventional simulation tools. However, articles such as those of Tooki *et al.* [11], Al-Saidi *et al.* [12], and Elsayed *et al.* [13] proposed improvements in reducing nonlinear effects, security, and signal coding, but without resorting to machine learning models. Additionally, the study by Wang *et al.* [14] was excluded because it focused exclusively on the design of temperature-sensitive physical components, without the direct application of artificial intelligence algorithms.

After selection, 23 articles were selected, which form the basis of this review and will be analyzed in the following sections.

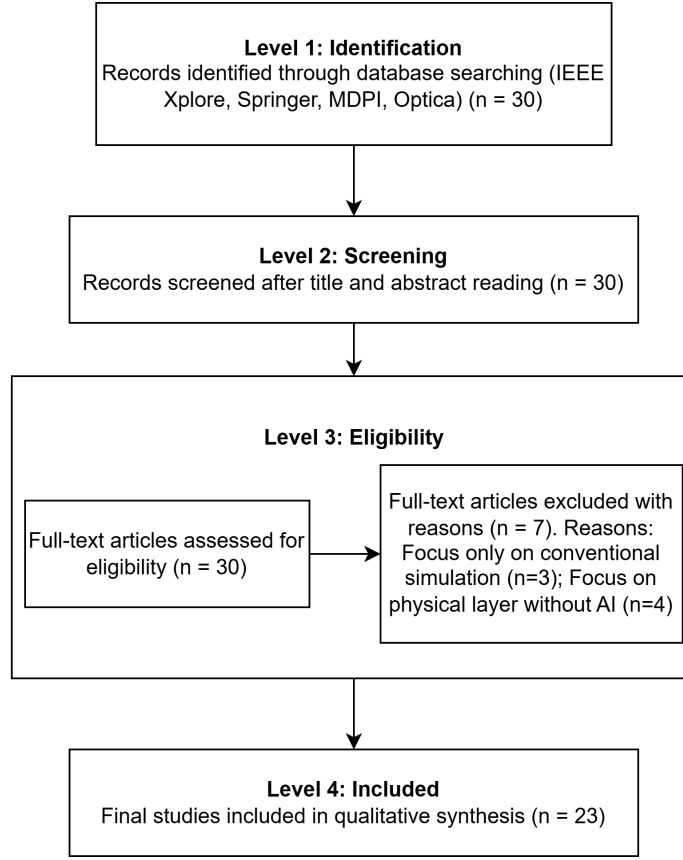


Figure 2. PRISMA flow diagram detailing the literature search, selection, and inclusion processes.

### III. CONCEPTS USED

To support the following analysis, this section briefly outlines the principles of optical networks and the artificial intelligence models referenced in the reviewed articles.

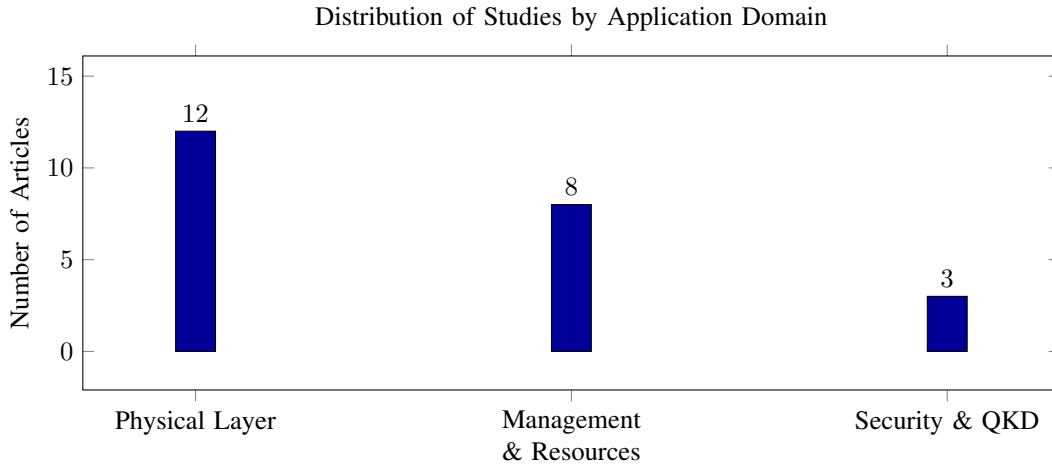


Figure 3. Quantitative analysis of the distribution of included references, organized by Artificial Intelligence application focus.

#### A. DWDM Networks and Physical Layer

The DWDM technology enables the transmission of multiple optical signals with different wavelengths over the same optical fiber. This approach increases the network's transmission capacity [3]. An optical infrastructure includes amplifiers, such as Erbium-Doped Fiber Amplifiers (EDFAs), to compensate for signal losses over distance, as well as equipment to route traffic [15].

However, the signal suffers degradation as it travels through the fiber due to physical impairments. Linear distortions include Chromatic Dispersion (CD) and Polarization Mode Dispersion (PMD), while nonlinear distortions, which present higher computational complexity to model, include effects such as FWM [4]. Inter-symbol interference (ISI) is also a problematic factor in high-speed links, impairing the QoT [3]. To evaluate signal quality at reception, performance parameters such as the bit error rate (Bit Error Rate (BER)) and the Q-factor [16], [17] are used.

### B. Machine Learning and Deep Learning

ML is an area of AI that allows computational systems to learn patterns from data without being explicitly programmed with fixed rules [18]. In optical networks, ML is used to solve problems that traditional analytical models either take a long time to compute or cannot solve accurately.

Classic ML algorithms, such as Support Vector Machines (SVMs), K-Nearest Neighbors (KNNs), and linear regression, are frequently used to quickly estimate QoT before establishing a new connection in the network [19], [20].

On the other hand, DL uses more sophisticated structures, such as Artificial Neural Networks (ANNs) and Recurrent Neural Networks (RNNs). These models are applied, for example, to compensate for severe nonlinear impairments, as they can map highly detailed relationships between the input and output of the optical channel [3].

### C. Network Management and Resource Allocation

Static planning and rule-based routing are no longer sufficient to handle the dynamic traffic patterns of 5G and 6G applications. While Software-Defined Networking (SDN) decoupling the control and data planes provided initial programmability [21], the growing complexity of multi-domain optical networks demands a more autonomous architecture. Consequently, current research focuses on Knowledge-Defined Networking (KDN). By embedding telemetry data and computational intelligence directly into the control plane, KDN drives the transition toward *zero-touch* operations and adaptive resource allocation [21], [22].

### D. Optical Security and Quantum Key Distribution

Integrating Quantum Key Distribution (QKD) into existing DWDM infrastructures is essential to secure data against emerging cryptographic threats. Among current methods, CV-QKD stands out because it generates secure keys using standard optical telecommunication hardware [23]. The primary challenge, however, lies in co-propagating these highly sensitive quantum channels alongside high-power classical data within the same fiber. This coexistence amplifies the impact of crosstalk, noise, and optical filter faults, making it difficult to distinguish between routine hardware malfunctions and malicious interception attempts [7], [23].

## IV. ARTIFICIAL INTELLIGENCE APPLICATIONS IN DWDM NETWORKS

The reviewed literature demonstrates that the implementation of AI algorithms in DWDM infrastructures has expanded across multiple network domains. The following analysis categorizes the principal solutions into three areas: physical-layer optimization, resource management, and security.

### A. Performance Optimization and Physical Impairment Mitigation

Accurate prediction of the QoT prior to establishing an optical connection is one of the most addressed topics in the literature. Traditional analytical models are considered computationally heavy, which is why several authors propose the use of ML to estimate QoT (such as Q-Factor, BER, and Optical Signal-to-Noise Ratio (OSNR)) quickly and based on data collected from the network [18], [22], [24]. Studies show that models such as ANNs and SVMs can predict the viability of optical signals with accuracies exceeding 90%, even when dealing with imbalanced or incomplete datasets [6], [19], [25]. Linear and nonlinear regression models have also shown effectiveness in predicting signal quality, decreasing the need for monitoring equipment [5], [16].

Beyond prediction, computational intelligence is applied to Optical Performance Monitoring (OPM) [26] and the mitigation of optical signal distortions. Malik *et al.* [27] propose a real-time State of Polarization (SOP) monitoring system to predict fiber damage before breakage occurs. In high-speed links, intersymbol interference (ISI) and nonlinear effects, such as FWM and Self-Phase Modulation (SPM), significantly degrade transmission. To reduce these impairments, Schädler *et al.* [3] proposed the use of RNNs for signal demodulation, compensating for nonlinear ISI more effectively than traditional equalizers. Additionally, ML techniques have been used to optimize the gain of hybrid optical amplifiers [15] and estimate their amplification spectrum [28], as well as to improve performance in systems combining complex modulation formats [4]. Optimization using ML also extends to Radio over Fiber (RoF) systems supported by DWDM, significantly improving BER and frequency allocation [17].

### B. Planning, Resource Allocation, and Management

Static planning and manual resource allocation are insufficient to handle the dynamic requirements of 5G and 6G networks. AI is therefore introduced into the control plane to automate decision-making. Venkatesan *et al.* [20] demonstrated that ML algorithms can be applied to the dynamic configuration of DWDM systems, automating efficient wavelength allocation. In the domain of dynamic reconfiguration, Prakash and Managuli [29] proposed using Reinforcement Learning (RL) to maximize network utilization and reduce Quality of Service (QoS) violations.

From a multi-layer management perspective, AI-based automation promotes seamless integration between the transport and access layers, optimizing real-time traffic routing [21]. This convergence is crucial to support the hybrid optical-wireless architecture required by 5G and 6G, in which AI and RL algorithms are applied to optimize transmission parameters in conjunction with radio resources, lowering latency and increasing throughput [1], [2]. Management also includes energy efficiency; Yurievna [30] highlights the use of AI and Big Data to manage power supply to network nodes, thereby promoting predictive maintenance and sustainability.

### C. Security and Resilience

In next-generation networks, the security of optical transmission is a critical factor. The introduction of AI has enabled the detection and diagnosis of traffic anomalies with high accuracy.

A particularly relevant case study focuses on QKD in DWDM networks. Traditional QKD systems shut down key generation whenever they detect channel degradation, assuming it to be an attack. However, optical filter malfunctions can cause the same type of situation. To avoid unnecessary downtime, Iqbal *et al.* [7], [23] developed ML algorithms, such as Deep Neural Networks (DNNs), capable of successfully distinguishing equipment failures from malicious attacks (e.g., power-splitting interception). The classification of these anomalies ensures not only continuous transmission security but also increased system availability.

Table I summarizes the main approaches and contributions identified in the literature. Thus, it is evident that the transition to autonomous optical networks relies on a wide range of algorithms, where ANNs and SVMs dominate the resolution of physical-layer problems (such as rapid QoT prediction and the mitigation of nonlinear effects). Conversely, RL and DNNs models assume a central role in dynamic multi-layer orchestration and the security of critical infrastructures, respectively. In sum, these studies demonstrate that computational intelligence efficiently mitigates the limitations of conventional approaches.

Table I  
AI APPLICATIONS IN DWDM NETWORKS

| Reference              | Focus / Domain            | AI Algorithm                | Overcome Challenge (Problem → Solution)                                                    |
|------------------------|---------------------------|-----------------------------|--------------------------------------------------------------------------------------------|
| Pal (2026) [1]         | Hybrid 5G/6G Networks     | ML, RL                      | <b>Static rules:</b> RL enables proactive optimization and dynamic allocation.             |
| Malik (2025) [27]      | Resilience / Self-healing | XGBoost                     | <b>Delayed reaction:</b> Continuous SOP monitoring predicts failures before they occur.    |
| Priyanshu (2025) [2]   | 6G / Edge Computing       | DRL, Federated Learning     | <b>High latency:</b> DRL reduces latency by 40% and saves 30% energy.                      |
| Prakash (2024) [29]    | DWDM Mesh / QoS           | RL, PSO                     | <b>Slow reconfiguration:</b> Adaptive global management increases network utilization.     |
| Assali (2024) [18]     | QoT Prediction            | ANN, DNN, Transfer Learning | <b>Slow analytical calculation:</b> Instant QoT prediction with >90% accuracy.             |
| Iqbal (2025) [23]      | CV-QKD / Security         | DNN, Random Forest, SVM     | <b>False positives:</b> Exact distinction between filter failure and hacker attack.        |
| Cichosz (2021) [6]     | QoT Prediction            | Isolation Forest            | <b>Lack of failure data:</b> Model trains only with normal data and detects anomalies.     |
| González (2025) [21]   | Multi-layer (IP/DWDM)     | GenAI, LLMs                 | <b>Manual operation:</b> GenAI translates business intents directly to SDN.                |
| Ibrahim (2025) [17]    | Radio-over-Fiber          | TensorFlow Regression       | <b>Crosstalk / FWM:</b> Automatic optimization of spacing to lower BER.                    |
| Kaur (2021) [15]       | Hybrid Amplifiers         | Bayesian Networks, HMM      | <b>Unequal gain:</b> Intelligent parameter adjustment for spectrum flattening.             |
| Yurievna (2025) [30]   | Energy Efficiency         | ML, Big Data                | <b>Electrical waste:</b> Predictive maintenance reduces node consumption by 28%.           |
| Kozdrowski (2021) [24] | DWDM QoT                  | Random Forest, XGBoost      | <b>Monitoring cost:</b> Prediction using only operator control data.                       |
| Morais (2018) [22]     | Unallocated Margin        | KNN, SVM, ANN               | <b>Margin waste:</b> Margin calculation in microseconds (error <0.4 dB).                   |
| Masih (2023) [16]      | Q-Factor / BER            | LASSO and Ridge Regression  | <b>Overfitting in ML:</b> LASSO defines nonlinear boundaries more accurately.              |
| Sakthivel (2025) [19]  | QoT Prediction            | SVM, LD-PCA                 | <b>Lack of generalization:</b> SVM reaches 99% accuracy for fast rerouting.                |
| Venkatesan (2022) [20] | Dynamic DWDM              | SVM, KNN, ANN               | <b>Slow provisioning:</b> Automation based on optical spectrum analyzer readings.          |
| Mujawar (2024) [25]    | Scalability / QoT         | QPSO-SASVM (Quantum)        | <b>Lack of convergence:</b> Quantum integration outperforms classical models.              |
| Lubana (2023) [28]     | Amplifier Gain            | ANN (LM and SCG)            | <b>Severe nonlinearities:</b> LM model achieves nearly perfect correlation (R=0.99).       |
| Kaur (2025) [4]        | OSNR Mitigation           | ANN, KNN, SVM               | <b>Destructive distortion:</b> KNN dominates OSNR prediction and SVM in noise calculation. |
| Jha (2021) [5]         | Transceiver Optimization  | MLP (TensorFlow)            | <b>Design complexity:</b> AI predicts Q-Factor using only BER figures.                     |
| Saif (2022) [26]       | Super-Channel             | ANN, Random Forest          | <b>High DSP cost:</b> Robust OSNR prediction using only Direct Detection.                  |
| Schädler (2021) [3]    | 800G long-haul DWDM       | Recurrent Networks (BRNN)   | <b>Severe ISI at 800Gbps:</b> BRNN outperforms conventional DSP at 600km.                  |

## V. OPEN CHALLENGES AND FUTURE TRENDS

Although the literature indicates promising results, large-scale implementation of AI algorithms in commercial DWDM networks still faces significant technical and operational obstacles. The analysis of the selected articles identifies the main current gaps and points out relevant directions for future research.

### A. Data Availability and Model Generalization

Optimizing ML models requires extensive and representative datasets. However, in operational optical networks, events such as signal degradation or critical failures are rare, leading to highly imbalanced and incomplete datasets [6]. Consequently, algorithms trained in simulation environments often exhibit generalization difficulties in real-world scenarios. Among future perspectives, the application of *One-Class Classification* classifiers and the exploration of *Transfer Learning* techniques stand out, as they allow adapting models to new domains or topologies with a reduced number of real samples.

### B. Interpretability and Trust (XAI)

The complexity of DL models often leads to a "black-box" nature, rendering them poorly interpretable. The literature indicates that telecommunications operators are hesitant to deploy non-interpretable models in networks with critical traffic. Consequently, the implementation of Explainable Artificial Intelligence (XAI) methodologies (such as SHAP and LIME) is shown to be fundamental to justify network predictions, ensure transparency in decision-making processes, and promote trust in infrastructure automation [2], [19].

### C. Computational Latency and Hardware Integration

Although DNNs achieves high accuracy in mitigating nonlinear distortions, the associated computational cost can be a limiting factor. Iqbal *et al.* [7] observe that the processing time of these networks can significantly exceed that of traditional probabilistic models, altering fast routing in dynamic scenarios. Future research is directed towards the execution and implementation of lighter models at the network edge [2] and, in the context of digital signal processing (Digital Signal Processing (DSP)), towards the direct implementation of models, such as RNNs, in Application-Specific Integrated Circuits (ASICs), thus reducing latency at the optical receiver [3].

### D. Knowledge-Defined Networking (KDN)

Transport network management is transitioning from the current SDN controllers to the KDN paradigm, whose objective is to achieve a "zero-touch" level of automation [22]. This new paradigm will increasingly use Generative AI and foundational models to convert business requirements, expressed in natural language by operators, into complex and precise network configurations [21]. Additionally, improving energy efficiency is critical; therefore, the use of AI will enable managing traffic load based on renewable energy availability, dynamically optimizing the energy consumption of optical nodes.

### E. Algorithm Security and Quantum Integration

As network management increasingly relies exclusively on computational methods, ML models themselves become susceptible to external attacks or data manipulation [17]. Designing architectures resilient to this type of cyberattack constitutes an ongoing challenge. Furthermore, sharing the DWDM infrastructure between traditional traffic and CV-QKD systems poses challenges for distinguishing between hardware failures and real attempts at cryptographic interception, requiring increasingly sensitive and rigorous classifiers to avoid compromising system uptime [7], [23].

## VI. CONCLUSION

Our work explores the real-world use of AI and ML in DWDM transport networks. After reviewing 23 selected studies, it is clear that these technologies have moved past mere theory [3]. They are now essential for handling the complexity and scaling limits that traditional analytical models often struggle with [1].

In the scope of optimizing the physical layer, it was demonstrated that algorithms such as ANNs and SVMs can predict QoT with accuracies exceeding 90%, significantly accelerating the establishment of optical channels. Furthermore, the application of networks such as RNNs proved to be highly effective in mitigating nonlinear distortions. Regarding resource management and allocation, AI enables dynamic orchestration, optimizing traffic routing and energy consumption, both of which are necessary for supporting the convergence of 5G and 6G architectures. Additionally, in the domain of security and resilience, AI proved crucial for distinguishing between hardware failures and cyberattacks, ensuring the continuous protection and availability of critical infrastructures, such as quantum channels (CV-QKD).

Although AI demonstrates high effectiveness in isolated domains of DWDM networks, the transition to autonomous *zero-touch* architectures under the KDN paradigm remains limited by the lack of interpretability of deep learning models and their computational latency. This review highlights the discrepancy between theoretical accuracy and real-time deployment viability. Consequently, subsequent studies will focus on integrating Explainable AI (XAI) frameworks and on developing hardware-accelerated algorithms for edge processing, aiming to ensure decision interpretability and reduce latency in optical infrastructures.

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